CS670 Assignment 2

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1 Part 1:

Given Documents in the dataset:

1. d1: The star in our solar system provides heat and light.
2. d2: That Hollywood star walked the red carpet for the movie premiere.
3. d3: Astronomers observe distant stars and galaxies using telescopes.

1.1 Preprocessing and Inverted Index:

1.1.1 Tokenization:

Output after performing tokenization on the documents:

1. d1: [The, star, in, our, solar, system, provides, heat, and, light, .]
2. d2: [That, Hollywood, star, walked, the, red, carpet, for, the, movie, premiere, .]
3. d3: [Astronomers, observe, distant, stars, and, galaxies, using, telescopes, .]

1.1.2 Stop Word Removal:

Output after removing stop words from the documents:

1. d1: [star, solar, system, provides, heat, light, .]
2. d2: [Hollywood, star, walked, red, carpet, movie, premiere, .]
3. d3: [Astronomers, observe, distant, stars, galaxies, using, telescopes, .]

1.1.3 Inverted Index:

1. star: d1, d2, d3 (lil doubt)
2. solar: d1
3. system: d1
4. provides: d1
5. heat: d1
6. light: d1
7. Hollywood: d2
8. walked: d2
9. red: d2
10. carpet: d2
11. movie: d2
12. premiere: d2
13. Astronomers: d3
14. observe: d3
15. distant: d3
16. galaxies: d3
17. using: d3
18. telescopes: d3

1.2 TF-IDF Document Matrix:

Term	Q	Counts (tf_i)			df_i	D/df_i	IDF_i	Weights ($w_i = tf_i \cdot IDF_i$)			
		D_1	D_2	D_3				Q	D_1	D_2	D_3
star	1	1	1	1	3	1	0	0	0	0	0
solar	0	1	0	0	1	3	0.4771	0	0.4771	0	0
system	0	1	0	0	1	3	0.4771	0	0.4771	0	0
provides	0	1	0	0	1	3	0.4771	0	0.4771	0	0
heat	0	1	0	0	1	3	0.4771	0	0.4771	0	0
light	1	1	0	0	1	3	0.4771	0.4771	0.4771	0	0
Hollywood	0	0	1	0	1	3	0.4771	0	0	0.4771	0
walked	0	0	1	0	1	3	0.4771	0	0	0.4771	0
red	0	0	1	0	1	3	0.4771	0	0	0.4771	0
carpet	0	0	1	0	1	3	0.4771	0	0	0.4771	0
movie	0	0	1	0	1	3	0.4771	0	0	0.4771	0
premiere	0	0	1	0	1	3	0.4771	0	0	0.4771	0
Astronomers	0	0	0	1	1	3	0.4771	0	0	0	0.4771
observe	0	0	0	1	1	3	0.4771	0	0	0	0.4771
distant	0	0	0	1	1	3	0.4771	0	0	0	0.4771
galaxies	0	0	0	1	1	3	0.4771	0	0	0	0.4771
using	0	0	0	1	1	3	0.4771	0	0	0	0.4771
telescopes	0	0	0	1	1	3	0.4771	0	0	0	0.4771

1.3 Boolen Retrieval:

Inverted index of star $\rightarrow$ [d1,d2,d3] and for light $\rightarrow$ [d1].
Only document d1 is common in both so that document is retrieved

1.4 Cosine Similarity and Ranking:

1. $|Q| = 0.4771$
2. $|d1| = 5 * \sqrt{(0.4771)^2} = 5 * 0.4771 = 2.3855$
3. $|d2| = 6 * \sqrt{(0.4771)^2} = 2.8626$
4. $|d3| = 6 * \sqrt{(0.4771)^2} = 2.8626$
5. $cos\theta_{d1} = \frac{Q.d1}{|Q||d1|} = \frac{04771*0.4771}{0.4771*5*0.4771} = 0.2$
6. $cos\theta_{d2} = \frac{Q.d2}{|Q||d2|} = 0$
7. $cos\theta_{d3} = \frac{Q.d3}{|Q||d3|} = 0$

- Final Ranking is $d1 > d2 = d3$
- Yes, Ranking is desirable as d1 would have been the most likely document we would have been searching for, given the query

1.5 Analysis of Word Sense Ambiguity:

1. In the query 'movie star', 'star' appears in all the document so the query reduces to movie alone. The word 'movie' appears only in document d2, so ideally, only d2 should be retrieved.

2. Using either Inverted Index (movie has only doc, d2 in its inverted index) or TF-IDF weights (d1 and d3 will have 0 cosine similarity with the query) will correctly retrieve d2.

3. In this corpus, star appears in all the documents, so system can not distinguish the meaning of star (has IDF=0).

4. System ignores it entirely in the retrieval of documents.

5. If the query was just 'star', all the docs will score equally and system would fail

6. Hence, Word sense ambiguity causes important query terms to lose discriminative power when they appear across multiple contexts.

2 Part 2: Building and Extending an IR System

All questions have been included in our submission

3 Part 3: Evaluating your IR System

3.1 Implementation

All questions have been included in our submission

Reported Values:

Metric	**Value @ k=10**
Precision@10	0.2831
Recall@10	0.4095
$F_{0.5}$-score@10	0.2887
AP@10	0.6586
nDCG@10	0.3620

Table 1: Evaluation Metrics at $k = 10$

Table 2: Evaluation Metrics on Cranfield Dataset

k	Precision	Recall	F-score	MAP	nDCG	MRR
1	0.6667	0.1115	0.3088	0.6667	0.3637	0.6667
2	0.5556	0.1811	0.3617	0.7133	0.3506	0.7133
3	0.5067	0.2405	0.3840	0.7222	0.3535	0.7356
4	0.4478	0.2750	0.3703	0.7146	0.3506	0.7422
5	0.3956	0.2978	0.3473	0.7085	0.3436	0.7467
6	0.3644	0.3283	0.3364	0.6996	0.3455	0.7519
7	0.3390	0.3505	0.3230	0.6858	0.3492	0.7519
8	0.3189	0.3740	0.3121	0.6754	0.3542	0.7524
9	0.3017	0.3974	0.3024	0.6639	0.3593	0.7539
10	0.2831	0.4095	0.2887	0.6586	0.3620	0.7539

3.2 MAP and MRR

3.3 Plotting evaluation metrics as a function of k

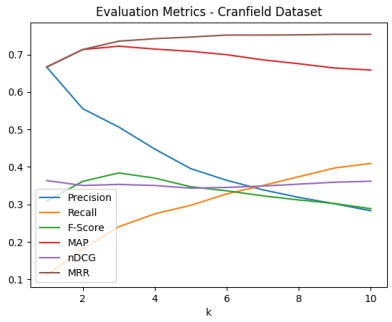


Figure 1: Trends of IR metrics

- **Precision decreases as k increases :** Precision is highest at lower ranks and gradually decreases as more documents are retrieved. This behavior is expected in most IR systems because the top-ranked documents are generally the most relevant, while lower-ranked documents tend to include more irrelevant results. The sharp drop from approximately 0.67 at k=1 to around 0.28 at k=10 indicates that the system is effective at identifying highly relevant documents early in the ranking.

- **Recall steadily increases with larger k:** Recall improves consistently as k increases because more relevant documents are retrieved when additional documents are considered. The increase from roughly 0.11 at k=1 to over 0.40 at k=10 demonstrates the expected trade-off between precision and recall in ranked retrieval systems.

- **F-score peaks at intermediate ranks:** The F-score initially increases and reaches its highest value around k=3 or k=4, after which it gradually declines. This suggests that the retrieval system achieves the best balance between precision and recall at moderate retrieval depths. Beyond this point, the decline in precision outweighs the gains in recall.
- **MAP remains consistently high:** The Mean Average Precision (MAP) curve stays relatively stable and high across all values of k, remaining above approximately 0.66 throughout the evaluation. This indicates that relevant documents are generally ranked near the top of the retrieved list, which reflects strong ranking quality.
- **nDCG remains stable across retrieval depths:** The nDCG values show only minor variation as k increases. This suggests that the system preserves good ranking order even when additional documents are retrieved. Since nDCG accounts for the positions of relevant documents, the stable trend indicates that highly relevant documents continue to appear near the top ranks.
- **MRR is consistently high:** The Mean Reciprocal Rank (MRR) curve remains very high and nearly constant across all values of k. This implies that the first relevant document is usually retrieved at very early ranks, often near the top position. Such behavior is particularly desirable in search systems where users typically focus on the first few retrieved documents.

The combined behavior of MAP, nDCG, and MRR indicates that the retrieval system is especially effective at early precision and top-ranked relevance. Even though precision decreases at higher k, the ranking quality remains strong because the most relevant documents are consistently prioritized near the beginning of the ranked list.

3.4 Run time

Total Runtime: 8.08431100845337 seconds

4 Part 4: Analysis

4.1 Limitations of the Vector Space Model

The vector space model represents documents and queries as vectors in a multi-dimensional space. This is effective for basic retrieval but has a lot of theoretical and practical limitations. In the basic vector space models the terms are considered as the dimensions and the weights in each dimensions are given by TF-IDF values. This can lead to the following problems:

1. **Synonymy**: Since VSM relies on exact word matches, it treats different words with the same meaning as an entirely distinct dimension. For example, a query involving aircraft will be very different from a query involving airplane since they are different terms. This can lead to false Negatives.

2. **Polysemy**: VSM cannot distinguish between different meanings of the same word. This can lead to false positives, unnecessarily retrieving irrelevant documents.

3. **Independence Assumption**: VSM assumes terms are independent of one another. Ideally the relation between terms should also be modelled.

4. **Ignoring Word Order and Context**: VSM ignores word order, complex phrases and the context in which a word appears. For example, stress on wings and wings under stress should be different things but are treated as the same thing.

5. **Lack of Semantic Understanding**: VSM does not capture the meaning of the documents/queries. No underlying concepts are captured.

4.1.1 Examples from the Cranfield Dataset

The following examples from the Cranfield collection demonstrate how the theoretical shortcomings of the VSM can be seen in actual retrieval:

- **Synonymy**: "Experimental investigation of the pressure distribution on an aerofoil"
  - This illustrates a failure to capture synonyms. A query using *"airfoil"* would fail to match this document, creating a false negative because the terms are mapped to distinct dimensions.

- **Polysemy**: "The effect of the pitch of the propeller on performance"
  - This sentence shows the risk of false positives. A query for *"aircraft pitch"* (referring to stability) would incorrectly retrieve this document because VSM cannot distinguish between propeller geometry and longitudinal rotation.

- **Independence Assumption**: "The study of boundary layer transition" The system treats *"boundary"* and *"layer"* as independent tokens. This allows irrelevant documents discussing "geological boundaries" or "atmospheric layers" to achieve high similarity scores.

- **Word Order**: VSM cannot differentiate the sentence
  "Interference effects of the wing on the body" from
  "Interference effects of the body on the wing."
  Both yield identical vectors, ignoring the specific aerodynamic context of which component influences the other.

- **Semantic Understanding**: "The analysis of thermal stress in structures"
  This would not be retrieved by a query for *'heat-induced deformation."* Since there is no keyword overlap, the system fails to bridge the semantic gap between the physical cause (heat) and the resulting effect (thermal stress).

4.2 The Zero-Result Problem (OOV Issue)

What happens when a query contains a word that does not appear in the document vocabulary (Out-of-Vocabulary term)?

- In VSM, the vocabulary is predefined by the unique terms indexed from the document during preprocessing. When an Out-of-Vocabulary term is encountered in a query, it has no corresponding entry in the inverted index.
- The system cannot compute a Term Frequency (TF) or Inverse Document Frequency (IDF) for that term. In the resulting query vector $\mathbf{q}$, the dimension corresponding to the OOV term is assigned a weight of zero.

How does this affect retrieval results in your system?

- Partial Match: If the query consists of both known and OOV terms, the OOV components are ignored. The cosine similarity is calculated using only the known terms, which may lead to a loss of query specificity.
- The Zero-Result Problem: If all terms in a query are OOV, the entire query vector $\mathbf{q}$ becomes a zero vector ($\mathbf{0}$).
- Mathematically, the cosine similarity between any document vector $\mathbf{d}$ and the query vector $\mathbf{q}$ becomes zero:

$$\text{Cosine Similarity} = \frac{\mathbf{d} \cdot \mathbf{q}}{\|\mathbf{d}\|\|\mathbf{q}\|} = \frac{\mathbf{d} \cdot \mathbf{0}}{\|\mathbf{d}\| \cdot 0} \to 0$$

- As a result, the system fails to retrieve any documents, even if relevant content exists within the corpus under different variations or synonyms.

Example query demonstrating this issue:

- The Cranfield dataset consists of aeronautics research mainly from the 1960s.
- Queries: "Effect of **nanoscale** coatings on **composite** wing durability." or "Using **GPU** acceleration for real-time flight **simulations**."
- Explanation: While the term "wing" is likely in the vocabulary, modern technical terms like **"nanoscale"** or **GPU** or specific contexts for **"composite"** materials may be OOV for this 1960s dataset. If these

key constraints are OOV, the system will either retrieve general "wing" or "flight" documents (ignoring the specific research intent) or return no results at all if the entire phrase is missing from the index.